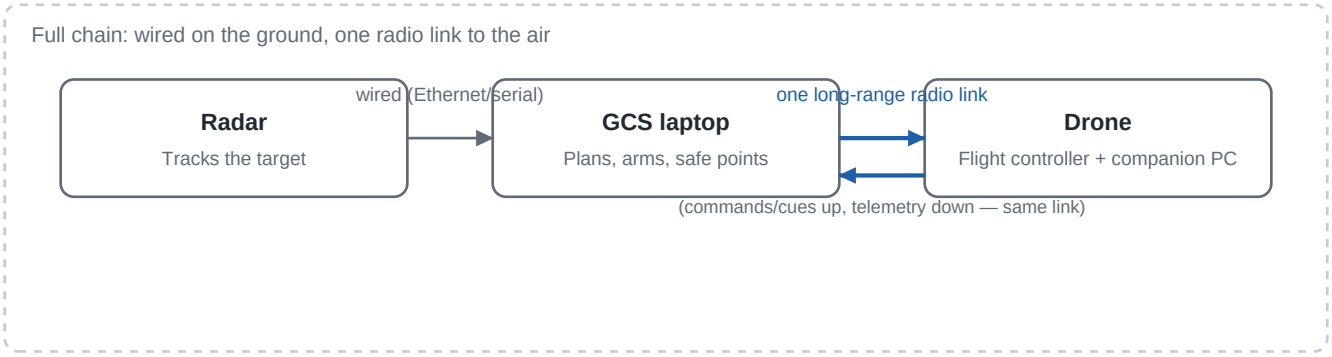
SKYFORCE Interceptor: Consolidated Design and Simulation Reference

Team 08, Anti-Drone Interceptor, Senior Track. Updated 10 October 2026, superseding the scattered sizing/resolution notes with one current baseline. Historic figures marked withdrawn below must not be mixed back in.

This is a preliminary engineering summary, not a flight approval. The simulation is a 2D point-mass, illustrative scenario model. Its outputs support geometric trajectory and state-logic discussion; they do not establish physical net capture, parachute deployment safety, or measured propulsion performance.

1. Radio chain: radar, GCS, and drone

This section answers directly what was previously stated inconsistently across documents: the drone has exactly one radio link to the ground, not two. "Ground radio" and "telemetry radio" in earlier documents are the same physical link, referred to by two different names. The radar's connection to the ground station is a separate, local, wired connection, not a radio link to the drone at all.



- Radar to GCS laptop: wired (Ethernet or serial), local to the ground segment. Not a radio link, and never reaches the drone directly.
- GCS laptop to drone: one bidirectional long-range radio link (RFD900x-class, 915 MHz, ~1 W, MAVLink). Target cues and commands travel up; GPS, battery, and status telemetry travel down. This is the item priced as "Long-range telemetry radio (air side)" in the bill of equipment — there is no second, separate "ground radio" item to buy.
- A manual RC safety-pilot override was considered and explicitly dropped by team decision (10 Oct 2026); the drone relies solely on this one radio link, so there is no RC transmitter/receiver in the current design.

2. Two-boundary launch gate (updated)

The keep-out geometry has been revised since the earlier sizing and architecture documents. The old 300 m illustrative keep-out radius is superseded — do not mix it with the current setting.

Boundary	Current value	Meaning
Outer tracking ring	2,000 m radius	Target enters this ring; begin track assessment, estimate position/speed/heading, predict path.
Inner protected boundary	1,000 m radius	Target should be intercepted before crossing this, if launch is authorised by the predicted path/time logic. Replaces the earlier 300 m placeholder.

Boundary	Current value	Meaning
Designated safe-point radius	120 m around each safe-point centre	Used after a geometric capture candidate, to decide immediate-release vs. tow. A separate concept from the 1,000 m boundary.

Launch gate logic (illustrative, constant-velocity track prediction — not a radar simulation):

- Predicted track misses the 1,000 m inner boundary → no takeoff for this predicted track.
- Predicted track enters the inner boundary and available time exceeds the 33.8 s response threshold → simulated launch recommendation.
- Predicted track enters the inner boundary but available time is at or below 33.8 s → no takeoff; too late under the current response-time assumption.

```
Response threshold = reaction delay + estimated intercept time + safety margin
                   = 1.8 s          + 30 s                   + 2 s
                   = 33.8 s   (illustrative hackathon demonstration value, not a certified launch rule)
```

What "fixed at one point" means: the interceptor has a fixed pad/base coordinate while awaiting the launch decision. The gate demonstration leaves it stationary and returns a simulated decision; the post-launch mission trajectories are separate cases that assume launch was already authorised. The code is not yet one continuous gate-to-landing simulation.

3. Current authoritative baseline

Parameter	Current value	Note
Configuration	Hexarotor, electric; intercept-only plus a separate towing branch	2D simulation is illustrative, not autopilot code
Motor	T-Motor U8 Lite KV100, 238 g each incl. cable	Published mass figure; verify exact supplier part
Propeller	T-Motor G28x9.2 CF, 28 in	Matches the cited motor test configuration; mass still to confirm
ESC	T-Motor Alpha 60A, 12S	Exact mass still to confirm
Max static thrust	7.3 kgf per motor, six motors	Published/static reference, not a forward-flight curve
Nominal / post-capture mass	18.00 kg / 23.50 kg	Use consistently in the simulation
Arm / frame span	0.800 m centre-to-motor; 2.31 m span	Coordinate with final prop clearance and CAD
Battery	12S, 8 Ah, 44.4 V nominal, estimated 1.97 kg	Mass estimated from generic energy density, not a chosen product yet

Mass ledger item	Mass	Meaning
Listed component subtotal	14.78 kg	
Integration / growth allowance	3.22 kg	Named, visible; not yet closed against real parts
Nominal mass	18.00 kg	14.78 + 3.22
Post-capture mass	23.50 kg	Used for the modelled tow branch

Before literal CAD freeze: confirm the propeller mass, ESC mass, the specific battery product and mass, and the net launcher + CO2 cartridge + net + parachute package envelope. The 3.22 kg allowance closes the arithmetic provisionally; it is not a substitute for a final itemised bill of materials.

Withdrawn first-pass values (do not reuse)

Old value	Status
U8 II KV100 with 22-inch propeller	Withdrawn
9.2 kgf max thrust per motor, ~41 A, ~1.965 kW	Withdrawn — did not match a published datasheet
4.0 kW sprint power	Withdrawn — current figure is 3.61 kW

Old value	Status
236 Wh intercept-only energy	Withdrawn — current ceilings are 207 Wh no-tow, 253 Wh tow
18.06 kg mass with a 2.50 kg unexplained gap	Replaced — 18.00 kg nominal, 3.22 kg named allowance, 14.78 kg subtotal
300 m keep-out circle	Superseded by the 1,000 m inner boundary (Section 2)

Thrust and model limits

Check	Nominal 18 kg	Post-capture 23.5 kg
Weight	176.6 N	230.5 N
Total static thrust (6 x 7.3 kgf)	429.7 N	429.7 N
Static thrust-to-weight ratio	2.43	1.86
Static-available acceleration estimate	21.8 m/s²	15.4 m/s²
Static thrust margin at 25 m/s	2.27x	1.79x
Margin at 25 m/s after assumed 30% de-rating	1.59x	1.25x

The 15/20/30% forward-flight de-rating figures are sensitivity assumptions, not measured data; 30% is the conservative headline case approved for the hackathon-level report. No measured thrust-vs-airspeed curve exists for the selected propeller. The 25 m/s speed cap and 4.5 m/s² acceleration cap are retained; the acceleration limiter must be applied uniformly in every phase, including mode transitions. A plotted max speed of 25 m/s shows the simulation stayed under its own configured cap — it is not, by itself, proof of measured flight capability.

4. Post-launch scenarios in the MATLAB file

#	Scenario	Inputs / logic	Expected outcome
1	Attack / slower target	Target [14, 5] m/s, ~14.87 m/s, tail-chase geometry	Geometric candidate in 5–15 m; immediate-release if inside a safe area
2	Defense / faster target	Target [27, 0] m/s, above the 25 m/s cap; lead-point geometry	Geometric candidate; demonstrates the lead-point concept, not physical capture
3	Tow required	Target [12, 0] m/s; candidate outside all 120 m safe-point radii	Tow to nearest safe point, <=60 s, 5 m/s setpoint
4	Abort	Target [-20, 0] m/s; abort if hard boundary crossed or time-to-boundary <= intercept time + 2 s	Pre-launch or in-simulation abort; no energy compliance scored if no launch occurred
5	PN diagnostic	Interceptor 20 m/s, target [0, 12] m/s, deliberate heading/LOS offset, N=3, 600 m blend range	Guidance diagnostic only, not a full mission case

Scenario 5 (PN diagnostic) is new since the last resolution note and directly answers the earlier concern that all three original scenarios started on a near-collision course and could not show PN correcting anything — this case deliberately offsets the initial heading to exercise that correction.

Capture-candidate and release/tow decision

Condition at candidate	Modelled decision	What it does not prove
Target mass <= 5 kg, inside any 120 m safe-point radius	Immediate-release state	Release mechanics or landing dynamics
Target mass <= 5 kg, outside all safe-point radii	Tow to nearest safe point, <= 60 s, low-speed waypoint control	Loaded flight, physical attachment, or actual controllability
Target mass > 5 kg at candidate	Simulation-only parachute-request flag; towing disabled (overrides the tow choice)	A real deployment command, or that the existing parachute can safely recover the load

Critical physical-design caveat: the 2.5 m parachute sizing (sizing document, Section 7) assumes a 5.5 kg suspended load (5 kg target plus an assumed 0.5 kg net/tether). A target above 6 kg would exceed that load case. The simulation flag represents intended state logic only — a rated parachute, load path, opening-shock and landing check would be needed before claiming safe recovery for an overweight target.

5. Tow transition correction, with the actual fault value now known

The earlier correction note (10 October) diagnosed this problem analytically, without a logged number. The change log now reports the actual previously observed spike: about 222.89 m/s² at the capture-to-tow mode transition — roughly 50 times the 4.5 m/s² cap, confirming the diagnosis (an abrupt, unramped speed-target change at the mode switch) rather than a minor rounding issue.

```
Documented fix: synchronise the mass update at capture; start the tow velocity target from the ACTUAL
capture-instant velocity (not an instant jump to cruise); ramp toward the 5 m/s tow setpoint; apply the same
4.5 m/s^2 limiter in every mode, including the transition itself.

Hand check (conservative 25 -> 5 m/s case, unchanged from the earlier note):
   dV = 20 m/s;  ramp time = 20 / 4.5 = 4.44 s;  average speed 15 m/s;  ramp distance = 66.7 m
```

This hand calculation shows the intended ramp logic is sound; it does not replace checking the logged acceleration in an actual rerun. That check is still open (Section 6).

6. Expected MATLAB outputs and verification checklist

Output file	Purpose
skyforce_simulation_summary.csv	One row per scenario: mode, gate status, target mass, candidate state, safe-area decision, tow status, speed/acceleration checks, energy estimates
skyforce_trajectory_export.csv	Sample-by-sample state log: time, phase, positions/velocities, LOS, range rate, closing speed, requested/applied acceleration, active limiter, power, cumulative energy
skyforce_propulsion_sensitivity.csv	Available thrust and margin for the 15/20/30% de-rating cases
skyforce_boundary_gate_summary.csv	Synthetic boundary scenarios: track entry, closest approach, time to inner boundary, response-time comparison, gate decision
skyforce_overweight_policy_summary.csv	Mass limit, parachute-request flag, towing allowed/not, deployment-modelled and recovery-validated fields

Minimum verification checklist for the next rerun:

- Boundary gate: confirm 2,000 m / 1,000 m are the radii actually reported, and check all three example decisions.
- Capture candidate: every candidate range lies in 5–15 m; post-capture positions must not overwrite the pre-capture closest-range metric.
- Safe area: immediate release only within a 120 m safe-point radius; otherwise tow, except the overweight case uses parachute-request/no-tow.
- Acceleration: inspect the maximum applied value for every phase, especially the exact sample at the mass/mode change — require <= 4.5 m/s², with no direct velocity clipping as a substitute.
- Closing speed: independently recompute from exported relative velocity and the LOS unit vector (V_close = – range rate).
- Energy: ensure the CSV run values, buffers, complete-mission estimate, and sizing ceiling all come from one run and one stated set of assumptions.
- Overweight case: verify the >5 kg flag and towing-disabled flag fire correctly; never label physical deployment as successful.
- File saving: confirm MATLAB printed the actual saved paths; filenames may be timestamped if a file was locked.

A PDF specification or hand calculation does not prove a particular MATLAB run passed. Use the latest local run and its exported CSV as evidence, and keep old CSVs/screenshots clearly versioned.

7. Energy estimate version history

Figure	Status
236 Wh no-tow	Withdrawn first-pass value
207 Wh no-tow / 253 Wh tow	Current authoritative sizing ceilings
39.55 / 41.44 / 103.53 Wh (attack / defense / tow)	Guide Decision Record estimates, using its then-current CSV values plus the agreed buffers
105.11 Wh (tow)	Towing-transition note's provisional estimate (84.73 Wh phase value + 1.58 Wh transition correction + 18.8 Wh buffer); explicitly not a corrected rerun
Current rerun value	Must come from the latest corrected MATLAB CSV — do not mix phase-energy values from different runs

Reporting rule: always display (1) energy used in simulated phases, (2) the added-phase buffer, (3) the resulting complete-mission estimate, and (4) the 207/253 Wh ceiling together. Label complete-mission values as estimates until one continuous launch-to-landing integration exists. The 20% demand margin and 80% usable-capacity convention are two distinct, intentionally stacked margins (established in the Guide Decision Record).

8. Applicability to fixed-wing or VTOL aircraft

Boundary and track-prediction logic (Section 2) is conceptually transferable: estimating target position, velocity and heading, predicting boundary entry, and checking available response time do not depend on the target or interceptor being a multirotor.

The current interceptor dynamics are not directly transferable. The 2D point-mass controller assumes multirotor-like speed control, acceleration limiting, the ability to slow to a low-speed towing target, and hold/waypoint behaviour — none of which a fixed-wing aircraft can do.

- Fixed-wing interceptor: would need a minimum safe airspeed / stall margin model, lift and drag, turn-radius and bank-angle limits, climb performance, and an explicit acknowledgement that it cannot hover or hold at a lead point.
- Hybrid VTOL interceptor: would need separate vertical-hover, transition, and wing-borne cruise dynamics, each with its own power and control limits.

9. CAD-freeze decisions, restated

- Arm geometry and span can proceed provisionally from the revised 28-inch propeller baseline.
- Before literal final freeze: confirm the propeller and ESC masses, the selected battery product and mass, and the net/CO2/parachute package envelope.
- Site radii and PN tuning are software/scenario settings, not airframe geometry, and do not block CAD.
- Physical capture and recovery performance remain intentionally out of scope for this preliminary deliverable.

Source record for this consolidation

This document consolidates: SKYFORCE Interceptor Sizing Calculations; MATLAB Intercept Simulation (What It Should Show); Resolution of Simulation Issues; Final Resolution of Remaining Simulation Issues; Guide Decision Record; Towing Transition Correction and Revised Energy Estimate; and the latest MATLAB source file. If a new run changes scenario energy or acceleration results, the current CSV export supersedes historical values here, with the change explained.